Building a Honeypot and Threat Intelligence Pipeline on AWS

Cowrie, Wazuh, n8n, AbuseIPDB, VirusTotal and TheHive

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Data period	18 to 25 October 2026, times as shown in the Wazuh dashboard

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1. Summary

I built a fake SSH server on Amazon Web Services and left it open to the internet. Real internet traffic found it. Some visitors only connected, and some logged in and acted like attackers. Every action they took was recorded, scored by my own rules, checked on AbuseIPDB and VirusTotal, and turned into an alert in TheHive.

What I achieved:

- A working honeypot lab with three servers, two separate networks and locked down firewalls.
- A pipeline that runs by itself: Cowrie, then Wazuh, then n8n, then TheHive and email.
- 152 connections from 83 different IP addresses between 18 and 25 October.
- 8 logins from 6 IP addresses. Three of those IP addresses uploaded a file after logging in, and VirusTotal flags all three files.
- One full case from the attacker 178.128.183.16: a weak password login, an uploaded Linux file that 46 of 64 security vendors flag as a miner, and a command to start it in the background.
- The alert reached TheHive, and I documented it as case 5 and closed it as True Positive.

This is a honeypot. Nothing real was infected, I never ran an attacker's file, and no IP address was blocked.

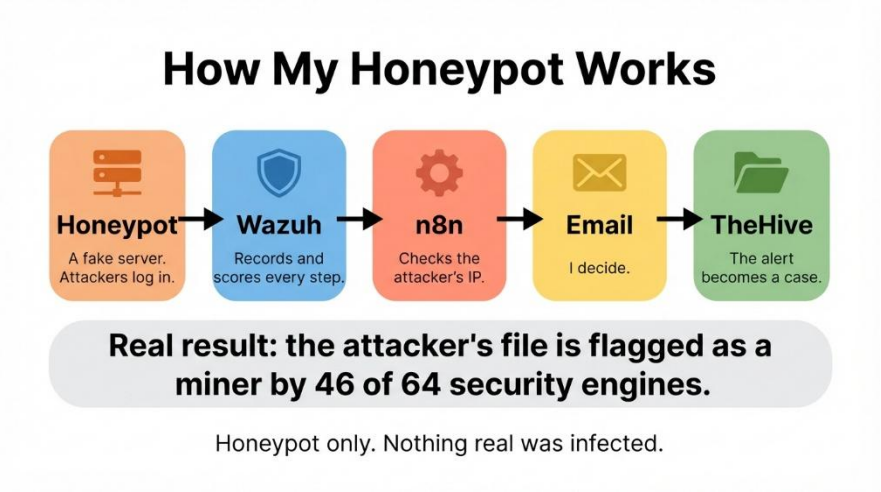


Figure 1:The whole project in five steps

2. The tools

Tool	What it is	Its job in my project
AWS	Cloud servers	Hosts everything in two separate networks
Cowrie	A honeypot that pretends to be an SSH server	Lets attackers log in and records every step
Wazuh 4.14.8	A system that collects logs and shows alerts	Applies my rules and gives each event a level from 3 to 12
n8n 2.43.7	An automation tool	Removes repeats, checks the IP, creates the alert, sends the email
AbuseIPDB	A site that scores IP addresses	Gives an abuse score from user reports
VirusTotal	A site that checks IPs and files	Shows how many security engines flag them
TheHive 5.8.1	A case management tool	Stores alerts and cases

3. Lab design

All servers run in AWS Mumbai on Ubuntu Server 24.04 LTS.

Server	Size	Job
Honeypot Server	t3.small, 20 GiB disk	Runs Cowrie in Docker and the Wazuh agent. This is the bait.
SOC Wazuh	m7i flex large, 60 GiB disk	Runs the Wazuh indexer, manager and dashboard
SOC Cases	m7i flex large	Runs TheHive and n8n in Docker

- The **SOC Network** (10.0.0.0/16) holds SOC Wazuh and SOC Cases. The **Honeypot Network** (10.1.0.0/16) holds the Honeypot Server. The two networks are not connected.
- Each server has its own Elastic IP, so its address stays the same after a restart.
- The only link from the honeypot to my SOC is two ports, 1514 and 1515, used to send logs to Wazuh.
- The SOC dashboards accept connections only from my own IP address.
- The honeypot accepts port 22 from anywhere. It can send traffic out only on port 443 and on the two Wazuh ports.

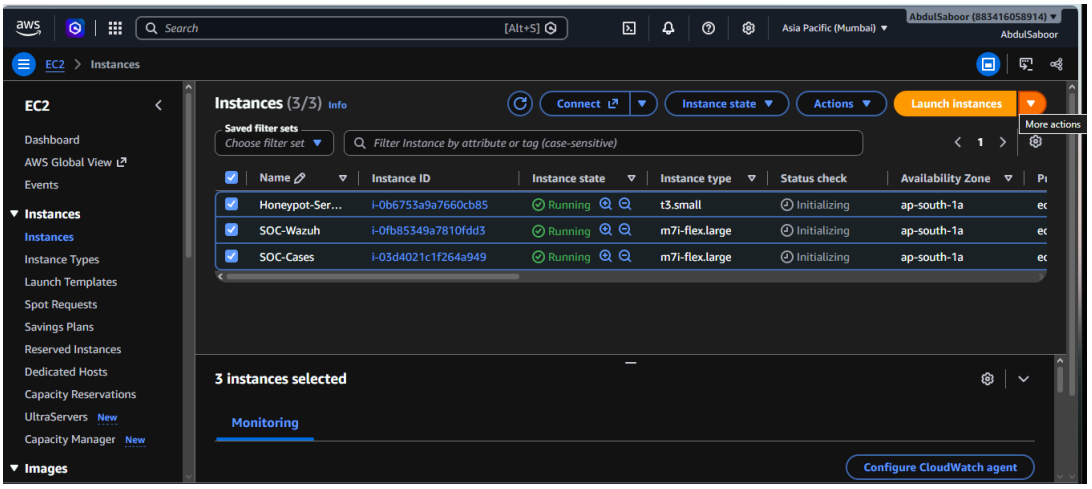


Figure 2: The three servers in AWS

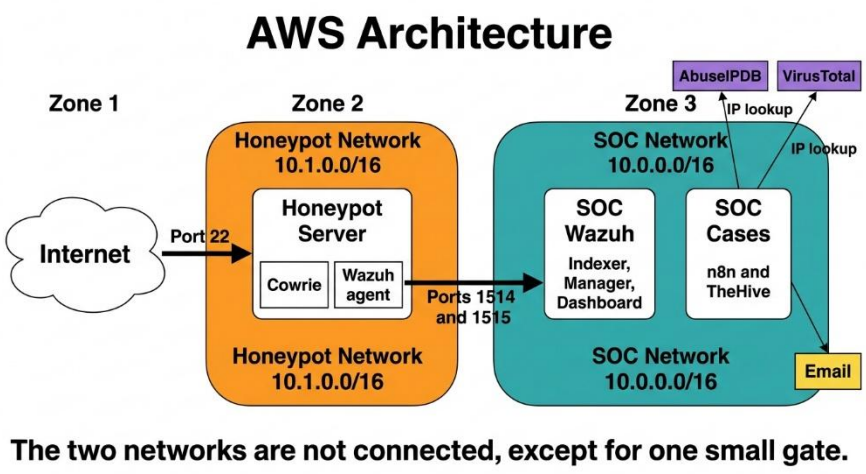


Figure 3:AWS architecture with two separate networks

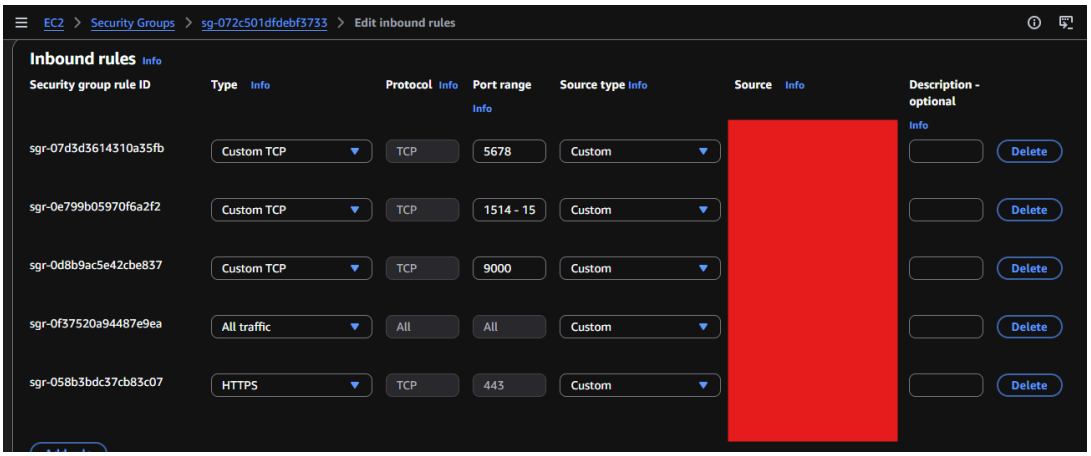


Figure 4:SOC firewall, inbound rules

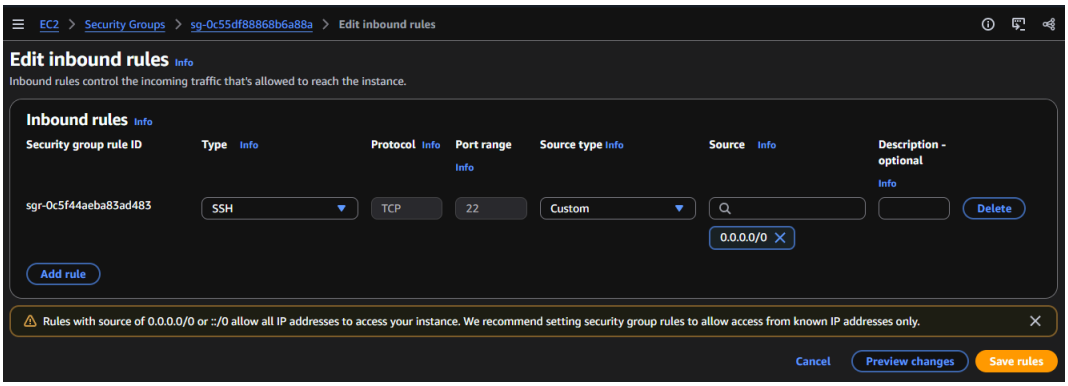


Figure 5; Honeypot firewall, only port 22 is open

4. How the pipeline works

1. An attacker connects to port 22. Cowrie answers like a real SSH server and writes every action to `cowrie.json`.
2. The Wazuh agent sends each line to the Wazuh manager.
3. My rules turn each event into an alert with a level from 3 to 12 (Appendix A). Wazuh adds the country.
4. Alerts of level 10 and above go to an n8n webhook.
5. n8n removes repeated events, then asks AbuseIPDB and VirusTotal about the IP address.
6. Important events create an alert in TheHive. The most serious ones also send me an approval email.
7. If I approve, the alert becomes a case. The workflow has no step that blocks an IP address.
8. A second n8n workflow sends me a daily summary of the honeypot.

The n8n workflow has these steps: Webhook, Edit Fields, Remove Duplicates, AbuseIPDB, VirusTotal, Summary, Dangerous?, TheHive Alert, Needs approval?, Send Email, Approved?, Promote to Case.

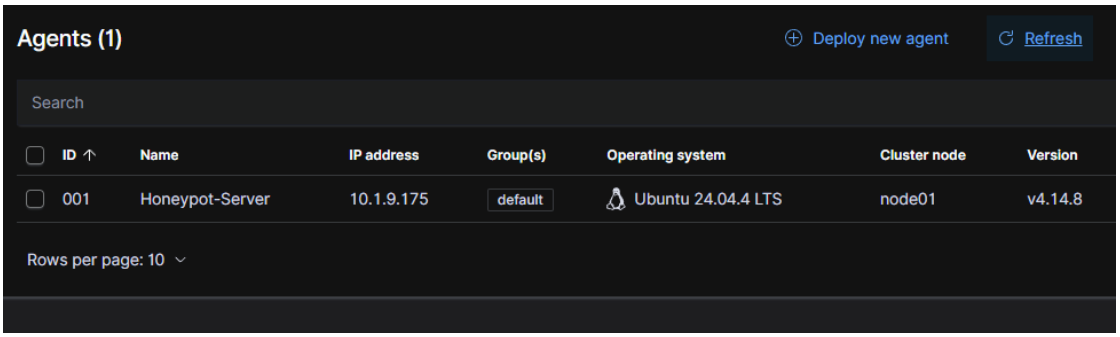


Figure 6:The honeypot agent is connected to Wazuh

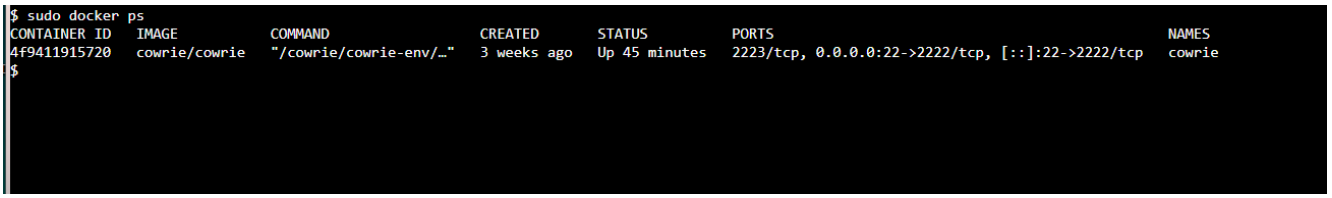


Figure 7:Cowrie running in Docker on port 22

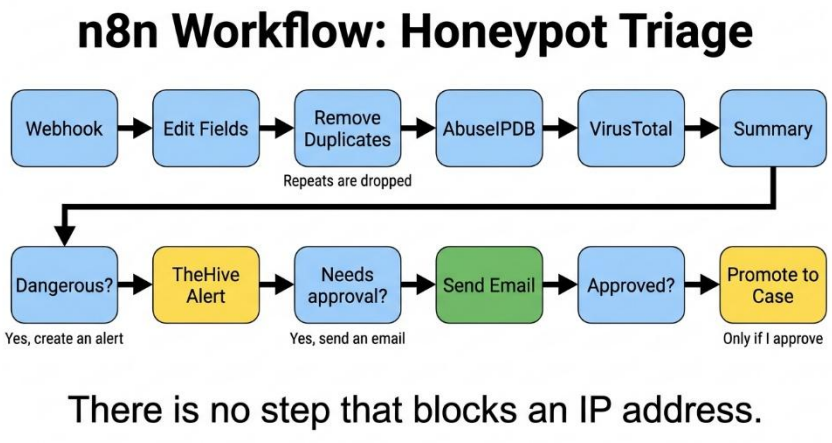


Figure 8:The n8n workflow as a diagram

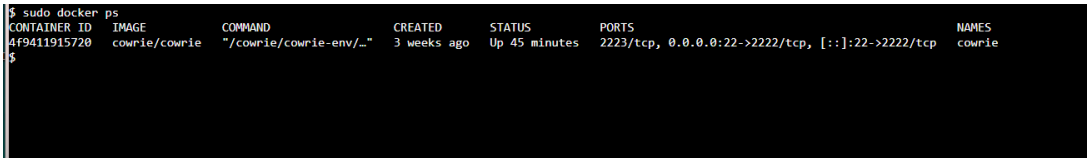


Figure 9:The Honeypot Triage workflow in n8n



Figure 10:A daily summary email

5. Safety

The honeypot only receives connections. I never opened or ran a file an attacker uploaded. I only looked up its SHA256 fingerprint on VirusTotal.

- The honeypot has its own network with no link to my other servers.
- The real SSH service on the honeypot server is switched off. I use the AWS browser terminal to work on it.
- The Docker container is not privileged and shares only its own volumes.
- On the honeypot server, the AWS metadata service is set to IMDSv2 only, so a program in the container cannot read the server's AWS role.
- I set a monthly cost budget with an email alert.
- I never type a real password into the honeypot, because it records every password.

6. Results

Period: I searched Wazuh from 12 October to 30 October. The honeypot's agent first started on 17 October, the first outside visitor came on 18 October, and the last one on 25 October at 16:17. So the search covers everything the honeypot recorded. My own test connections are excluded.

Measure	**Value**
Connections	152
Different IP addresses	83
Successful logins	8, from 6 IP addresses
Failed logins	2
IP addresses that uploaded a file after login	3
IP addresses that ran the same test script	2 (Romania)

Most visitors only connected and left within seconds. I read these as internet scanners.

Of the 83 IP addresses, 76 only connected and never tried a login. The other 7 tried to log in or did log in.

All 8 logins

#	Time	IP address	Country	Login
1	21 Oct, 12:21	34.79.89.168	not checked	admin, admin
2	23 Oct, 23:44	202.104.14.68	China	root, debian
3	24 Oct, 05:15	92.118.39.71	Romania	root, !root
4	24 Oct, 05:17	92.118.39.71	Romania	root, 111111
5	24 Oct, 05:19	92.118.39.71	Romania	root, 123123
6	24 Oct, 09:00	166.88.33.16	United States	root, ubuntu
7	24 Oct, 21:42	92.118.39.77	Romania	root, !root
8	25 Oct, 14:02	178.128.183.16	United States	root, ubuntu

Every login used a default or weak password.

The countries in this table come from the TheHive alert tags. Wazuh shows a country only for events after 24 October at 16:20, when I added the location processor.

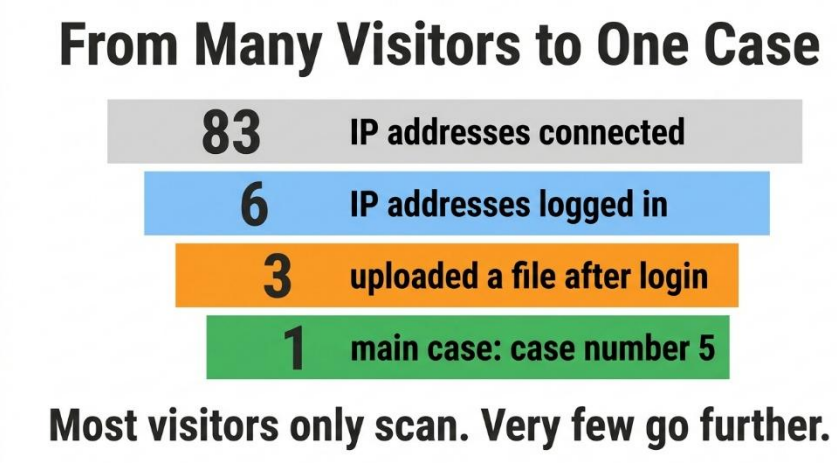


Figure 11: From many visitors to one main case

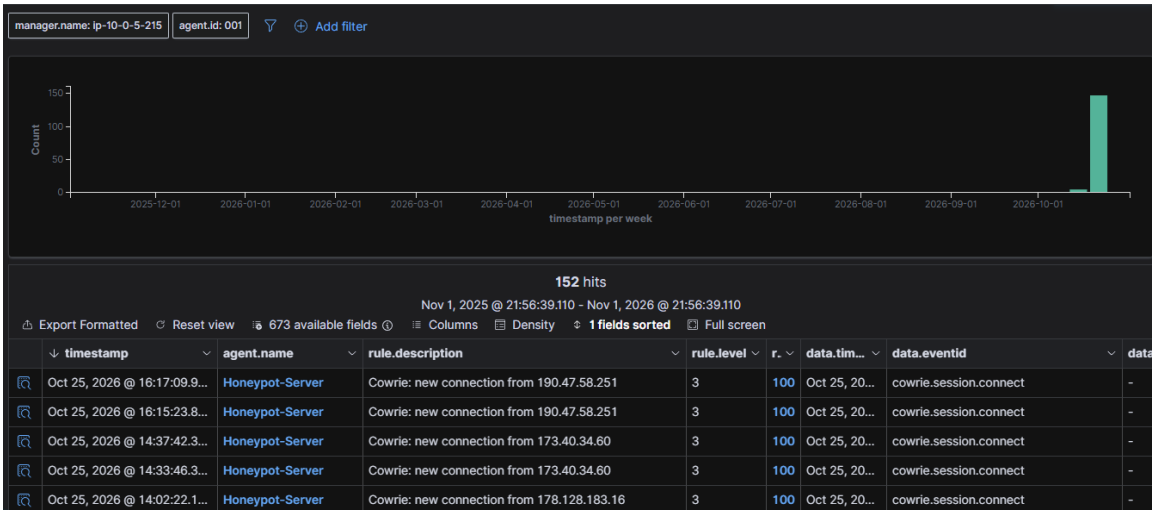


Figure 12:Wazuh events: 152 connections

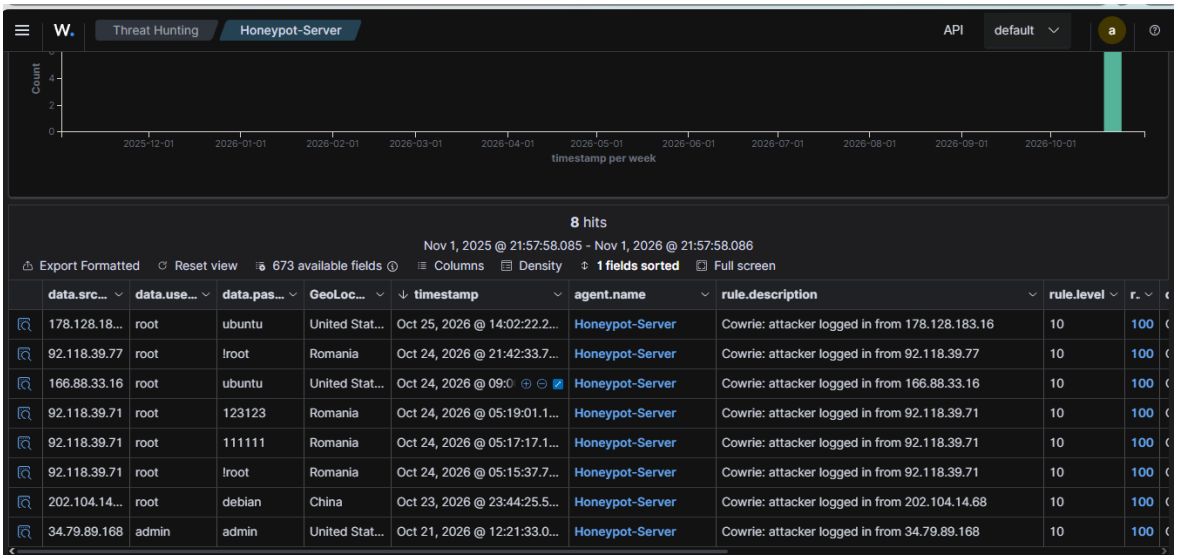


Figure 13:All successful logins

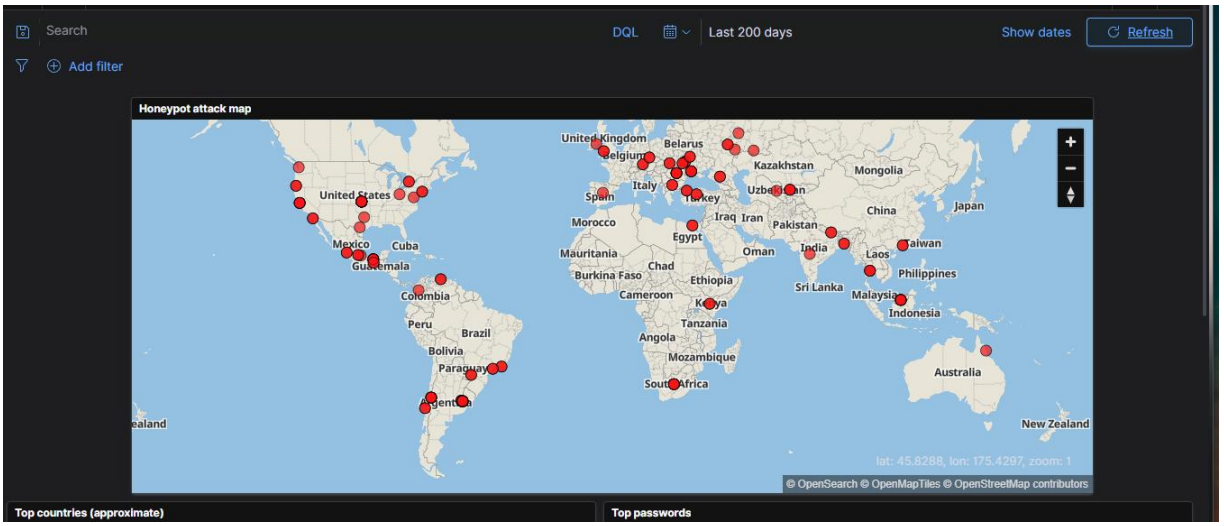


Figure 14:connections by country

7. Main case: attacker 178.128.183.16

7.1 Who it was

Item	Value
IP address	178.128.183.16
Location	United States, Santa Clara, California
Network	DigitalOcean, LLC, AS14061 (data center and web hosting)
Session	0f14d801afa9, about 4 minutes 26 seconds
Login	root with the password ubuntu, accepted on the first try
Behaviour	No failed logins and a fixed pattern. It looks automated. This is my reading.

7.2 What it did, on 25 October

Time	What happened	Rule	Level
14:02:20	New SSH connection	100101	3
14:02:22	Logged in as root with the password ubuntu	100103	10
14:02:24	My n8n workflow started	none	none
14:06:45	Uploaded a 28.90 MB file named sshd	100100	3
14:06:46	Ran a command to start it	100104	10
14:06:46	Session closed	100100	3

These are the times in Cowrie's own log. Wazuh shows the same events up to two seconds later.

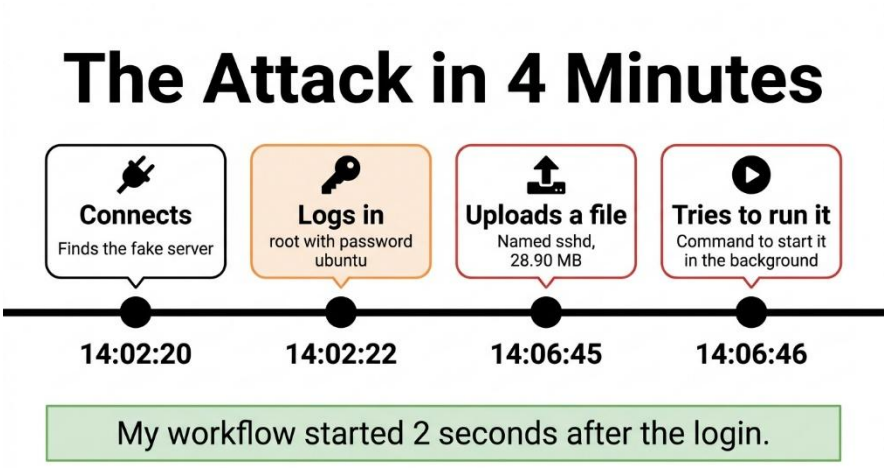


Figure 15:The attack in four minutes

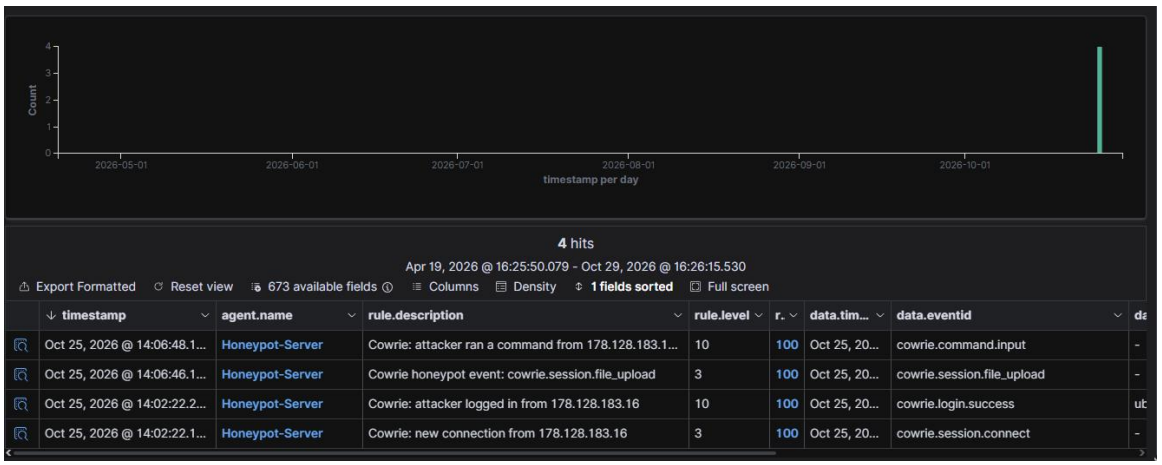


Figure 16:The four key events of the session

7.3 The command

chmod +x ./.908602877287403466 5/sshd; nohup ./.908602877287403466 5/sshd <46 IP addresses> &

Part	What it means	Why it looks suspicious
./.9086028772874034665/	A folder whose name starts with a dot	A hidden folder
sshd	The file name	Copies the name of the real SSH service
chmod +x	Makes the file runnable	Activates the uploaded file
nohup	Keeps a program running after logout	It keeps running after the attacker leaves
&	Runs it in the background	It stays quiet
46 IP addresses	Extra information given to the program	It looks like a peer list for a mining botnet. This is my reading.

I do not print the 46 addresses, because some may belong to ordinary infected computers. The full command is saved in my Wazuh logs. Cowrie only plays commands, so the file never really ran.

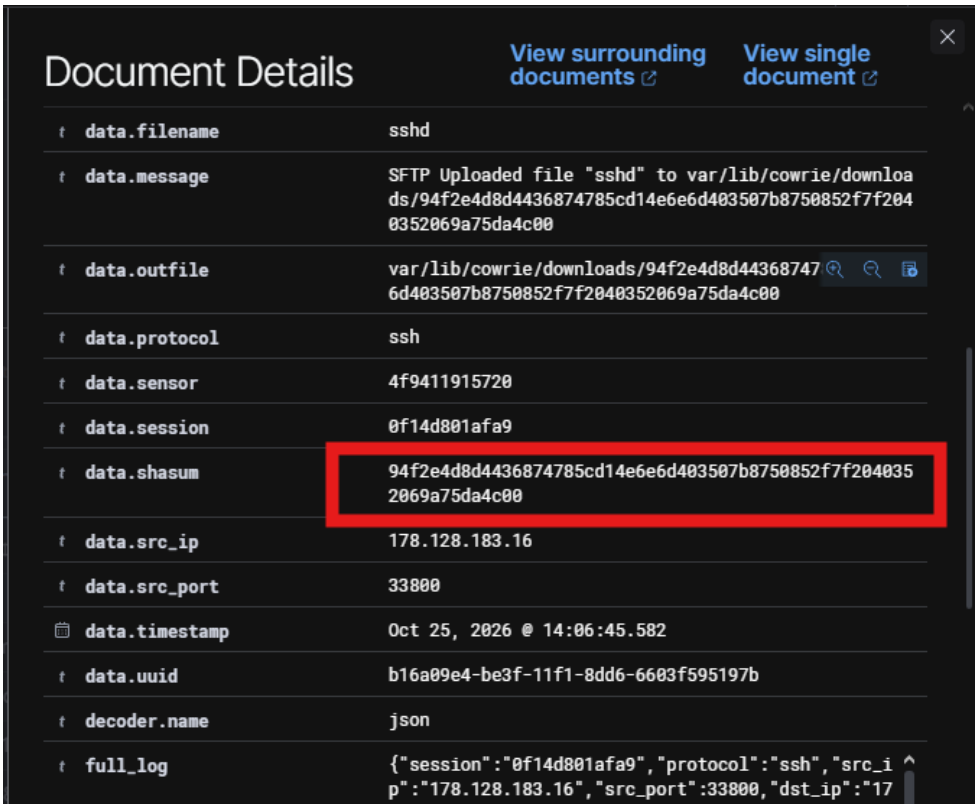


Figure 17:The upload event with the file name and SHA256

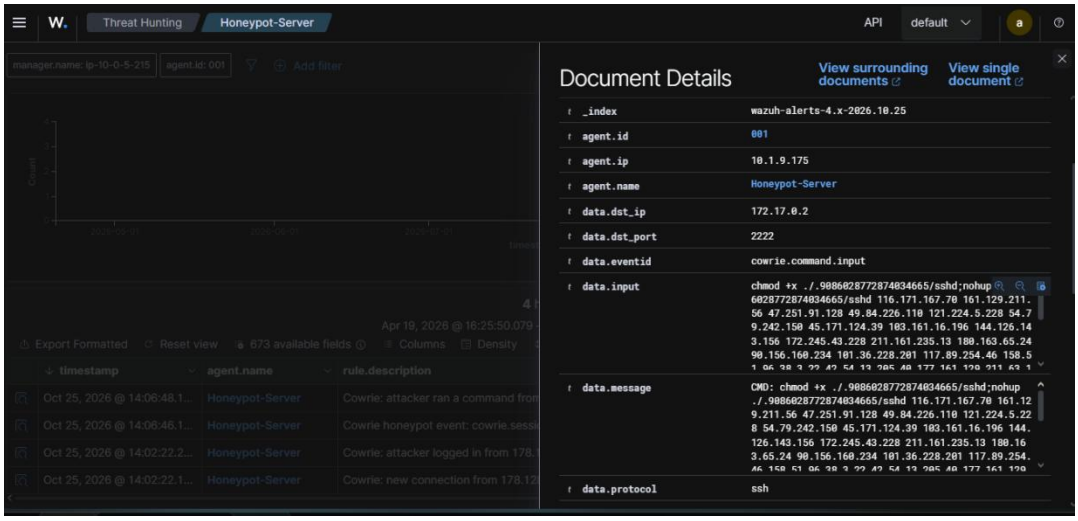


Figure 18:The command event

7.4 The uploaded file

Item	Value
SHA256	94f2e4d8d4436874785cd14e6e6d403507b8750852f7f2040352069a75da4c00
Type and size	ELF 64 bit Linux file, 28.90 MB
VirusTotal	46 of 64 vendors flag it as malicious
Threat label	miner.genericrxss/multiverze (categories miner, trojan, pua)
Kaspersky	HEUR:Trojan.Linux.Miner.gen
ESET	Linux/CoinMiner.ABF Trojan
Microsoft	Trojan:Linux/Multiverze!rfn
Sophos	Linux/Miner-AGY

VirusTotal uses the words “miner” and “CoinMiner”, so I call the file a cryptominer. I did not analyse the file myself.

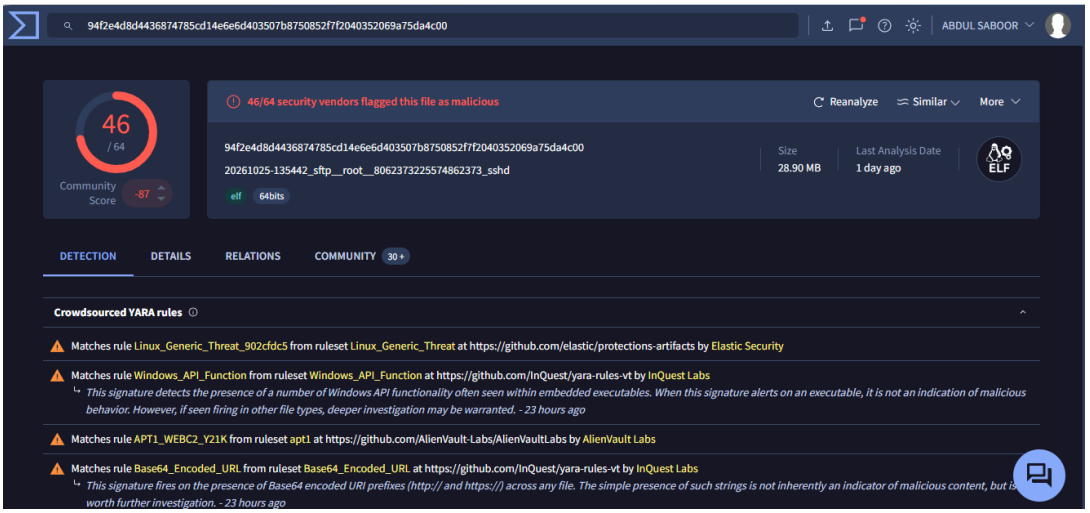


Figure 19:VirusTotal result for the uploaded file

7.5 What the IP looks like online

Source	At alert time (25 October)	When I checked (29 October)
AbuseIPDB	Score 100, 1,127 reports	100 percent, 1,662 reports from 591 sources
VirusTotal (IP)	10 engines	9 of 92 vendors

The numbers grow and change over time. Both sets are real. AbuseIPDB shows the IP was first reported on 5 January 2026, and the latest report came a few hours before I checked, so it is still active. It was reported for SSH brute force, port scans and hacking, on ports including 22, 23, 445, 1433, 3306, 3389, 5900 and 8080.

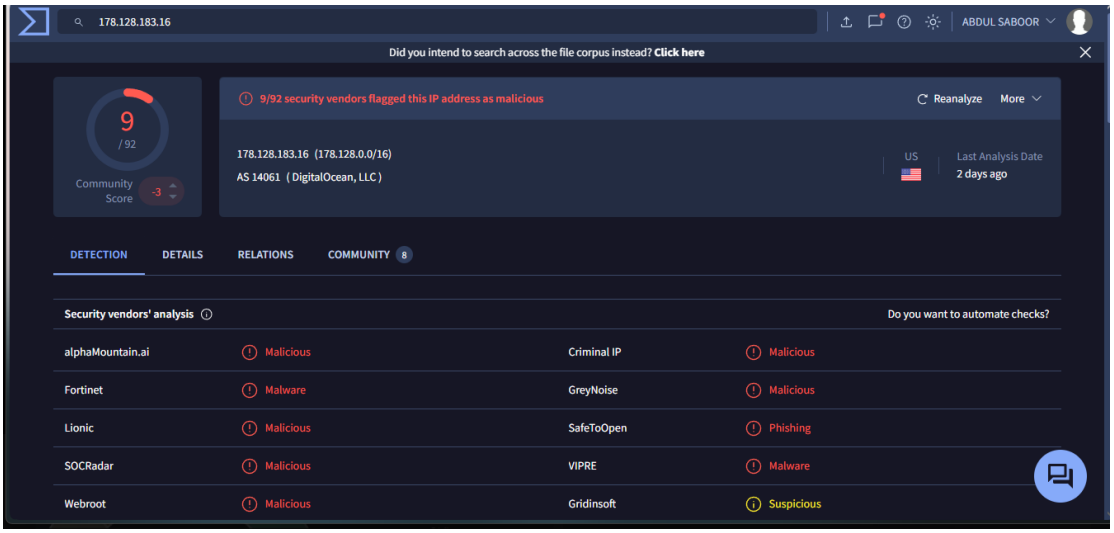


Figure 20:VirusTotal result for the attacker IP

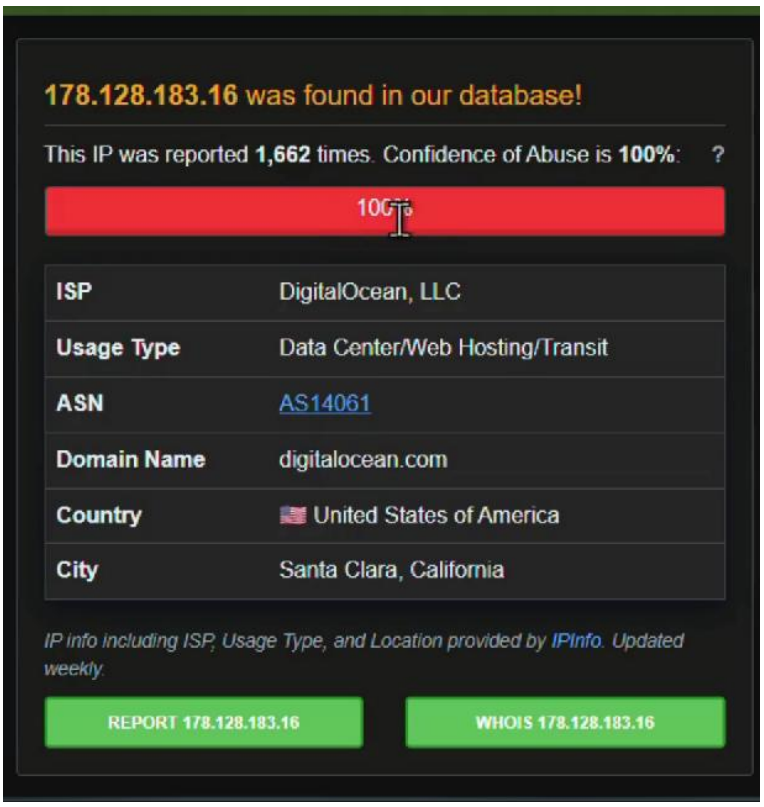


Figure 21:AbuseIPDB result for the attacker IP

7.6 How my pipeline responded

- Wazuh raised a level 10 alert at the login, and my n8n workflow started two seconds later.
- n8n checked the IP and created the TheHive alert “Cowrie: attacker logged in from 178.128.183.16” (severity High, one observable). It also sent me the email “Honeypot attacker 178.128.183.16 needs approval”.
- I did not answer the approval email in time, so I created case 5 from the alert myself on 29 October 2026 at 08:07. I added the upload and the command to the description, started the case and closed it as True Positive with the impact set to none, because it is a honeypot.
- No IP address was blocked.



Figure 22:n8n runs for 25 October

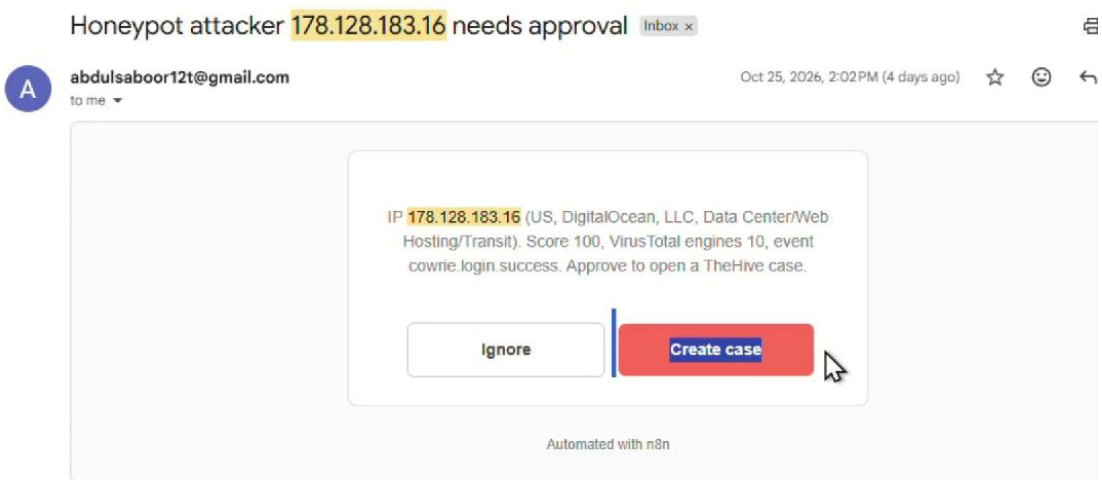


Figure 23:The approval email

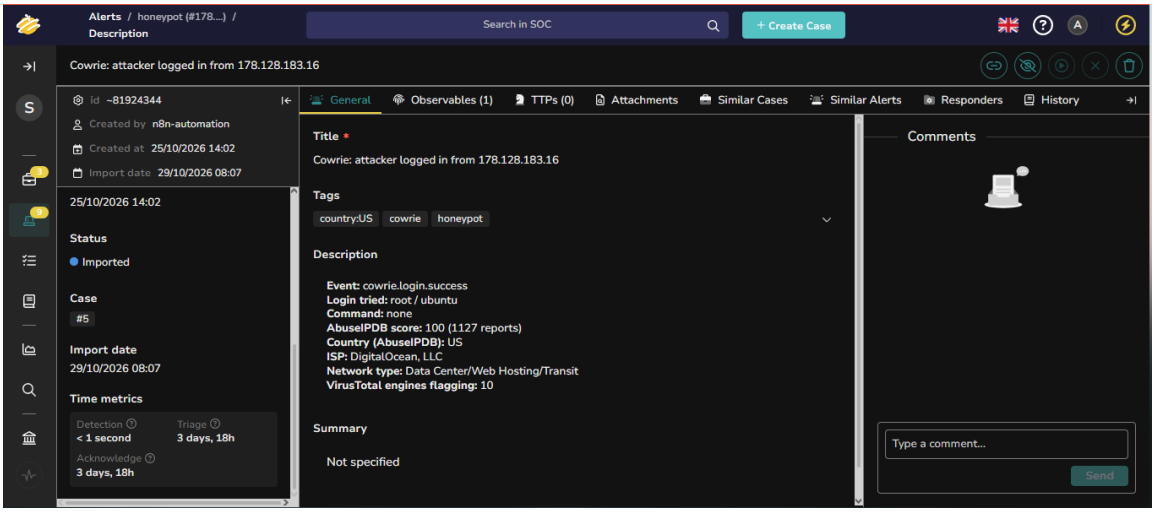


Figure 24:The TheHive alert

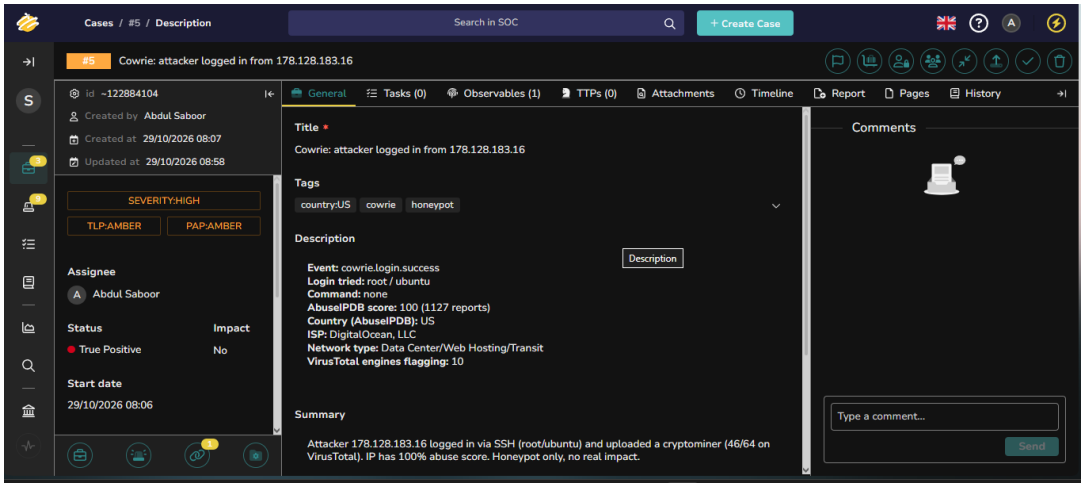


Figure 25:TheHive case 5

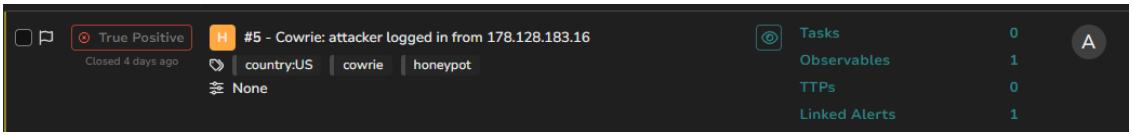


Figure 26:Case 5 closed as True Positive

8. Other attackers

Three uploads after login

Three IP addresses logged in and then uploaded a file. For two of them the upload came almost exactly five minutes after the login. For the third (the main case) it came after 4 minutes and 23 seconds.

IP address	Country	Login	Upload	SHA256 of the file	VirusTotal
202.104.14.68	China	23 Oct, 23:44	23 Oct, 23:49	3f48195796d3080ddac3da83346d8da802f9bf34eecba0be0ebbad3c78c267c1	42 of 65 vendors
166.88.33.16	United States	24 Oct, 09:00	24 Oct, 09:05	776c0fccae481c384f5636903ce500fae22803cf8778ec9a029956bb85a68010	36 of 62 vendors
178.128.183.16	United States	25 Oct, 14:02	25 Oct, 14:06	94f2e4d8d4436874785cd14e6e6d403507b8750852f7f2040352069a75da4c00	46 of 64 vendors

All three files have different fingerprints, and many vendors flag all three. I analysed only the third one in detail. VirusTotal shows other file names for these files, but those names come from other people who submitted them, so I do not use them as evidence.

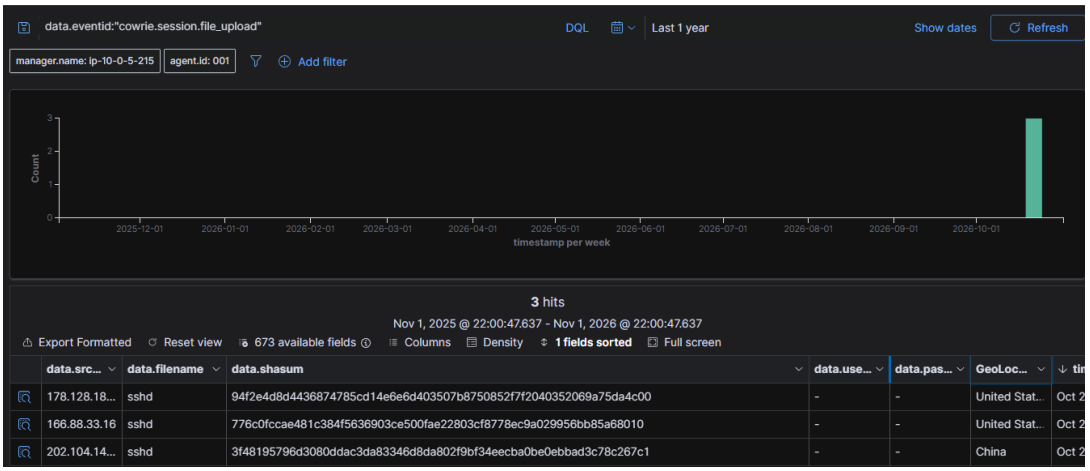


Figure 27:The three upload events

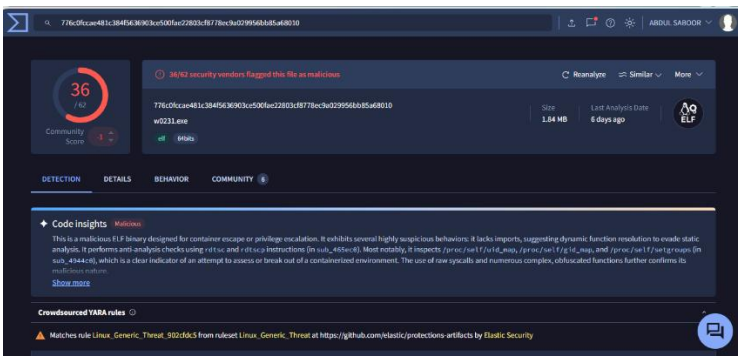


Figure 28:VirusTotal result for the file from 166.88.33.16

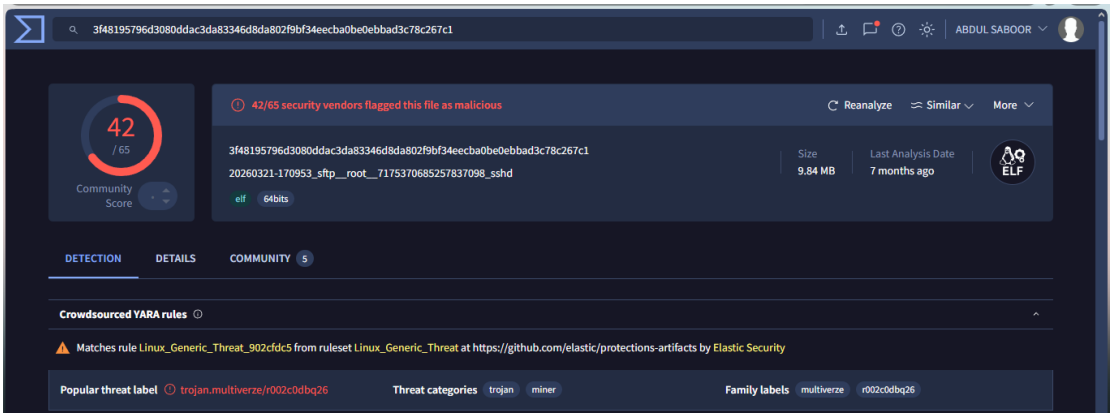


Figure 29:VirusTotal result for the file from 202.104.14.68

Two IP addresses in Romania

The IP addresses 92.118.39.71 and 92.118.39.77 both belong to the hosting company DMZHOST. They logged in as root with the password !root, and 92.118.39.71 also used 111111 and 123123. Each session first ran a long script that collects the system's name, CPU, graphics card and recent logins, and then a tiny script that creates a file named filter, runs it and deletes it.

The filter file has the SHA256 af77b643964afd794460d28ab8a11b0ec8790cd5463abfde126613a4f3bccd32, and no VirusTotal vendor flags it. The same fingerprint appeared in all four sessions, three from 92.118.39.71 and one from 92.118.39.77. My reading is that the bot tests whether the machine is a real computer or a honeypot. Both IP addresses ran the same routine from the same hosting company, so I think they belong to one campaign. TheHive case 4 is for 92.118.39.71.

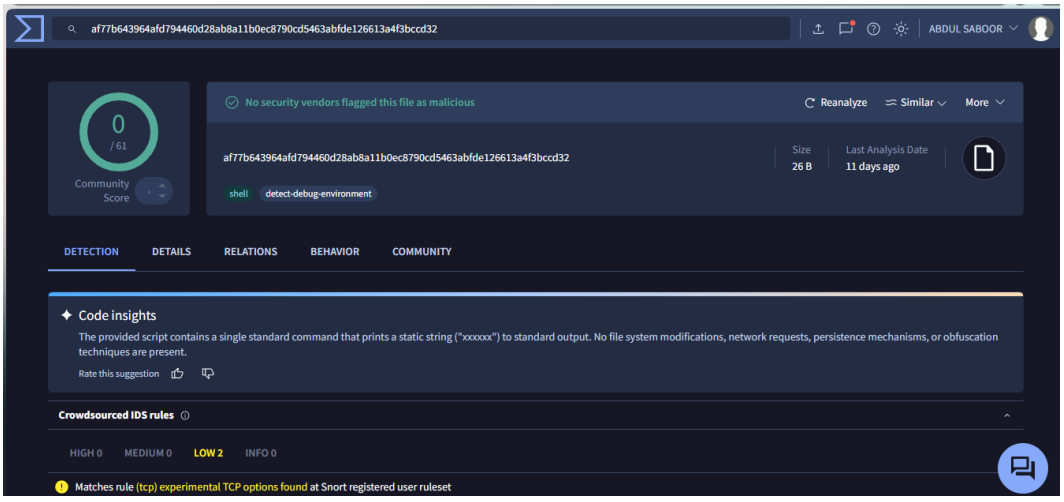


Figure 30:VirusTotal result for the filter script

9. What I built and learned

Challenges I solved during the build:

- Got Wazuh installed after two failed attempts caused by a package lock and a half removed package.
- Made Wazuh show a country for each attacker, by adding a Filebeat processor for Cowrie's field name `src_ip`.
- Wrote my own Wazuh rules from the real field names, using the Ruleset Test page.
- Fitted TheHive and Cassandra into an 8 GiB server by lowering their memory limits.
- Connected n8n to TheHive and the Wazuh indexer, fixing JSON and request method errors.

Skills I practised: cloud networking and firewalls, log collection, detection rules, automation, threat intelligence lookups, case management and incident documentation.

10. MITRE ATT&CK and indicators (main attacker)

My own assessment

Technique	Evidence
T1021.004 Remote Services: SSH	SSH connection to port 22
T1078 Valid Accounts	Login as root with the password ubuntu
T1105 Ingress Tool Transfer	File uploaded over the SSH connection
T1222.002 File Permissions Modification	chmod +x
T1564.001 Hidden Files and Directories	Folder name starts with a dot
T1036.005 Masquerading	File named sshd
T1496 Resource Hijacking	VirusTotal label: miner

Indicators of compromise

Type	Value
IP address	178.128.183.16 (DigitalOcean, AS14061)
SHA256	94f2e4d8d4436874785cd14e6e6d403507b8750852f7f2040352069a75da4c00
File name	sshd, 28.90 MB, ELF 64 bit
Hidden folder	./.9086028772874034665/
Login	root with the password ubuntu
Other upload fingerprints seen on my honeypot	776c0fccae481c384f5636903ce500fae22803cf8778ec9a029956bb85a68010 (166.88.33.16), 3f48195796d3080ddac3da83346d8da802f9bf34eecba0be0ebbad3c78c267c1 (202.104.14.68)
Test script file named filter	af77b643964afd794460d28ab8a11b0ec8790cd5463abfde126613a4f3bccd32

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11. Recommendations

For real servers: use SSH keys only, do not allow root login, use fail2ban against repeated logins, allow SSH only from known IP addresses, watch outbound connections, and use an EDR tool to catch unknown files named like system services.

12. Limits and next steps

- The honeypot recorded visitors for one week, and it was not running all the time.
- Cowrie only plays commands, so I cannot say what an uploaded file would really do. The words “miner” and “botnet” are my reading of VirusTotal’s labels and the IP list.
- AbuseIPDB and VirusTotal numbers change over time.
- Case 5 was created by me from the alert, four days after the attack.
- An IP address is not a person. Many of these addresses belong to rented servers, so I describe behaviour and do not accuse anyone.

Next steps:

1. Add a Wazuh rule at level 12 for file uploads. Today they are level 3 and make no alert.
2. Fix the n8n run for the attacker’s command, which ended with an error.
3. Give the approval email a longer wait and a reminder.
4. Run the honeypot for several weeks and compare the results.

13. Conclusion

I built a complete honeypot and detection pipeline on AWS. It recorded real internet visitors, including attackers, scored their actions, checked them on outside services and created alerts and a case in TheHive. In my main case, an attacker logged in with a weak password, uploaded a file that 46 of 64 security vendors flag as a miner, and tried to run it, all within four minutes. My pipeline started two seconds after the login. The project shows that a small lab can find real threats, and it gave me a clear list of improvements for the next version.

14.Disclaimer

This project was conducted solely for educational, research, and defensive cybersecurity purposes. No unauthorized attacks, exploitation, or access to third-party systems were performed. The honeypot was used only to observe and record unsolicited internet traffic. Malicious files were not executed, and no IP addresses were blocked or attacked. The information in this report must not be used for illegal or harmful activities.